

Milestone 2: The Longevity Equation

1. Project Goal

The goal of our project is to provide a multi-dimensional perspective on global life expectancy. We want to enable users to explore how a country's wealth, gender demographics, and chronic disease rates impact the lifespan of its citizens. By visualizing the data interactively, we give users the opportunity to discover patterns, such as why certain high-income nations fail to achieve proportional health outcomes.

Our visualizations are powered by a consolidated dataset bridging World Bank and WHO records from 2000 to 2021, totaling over 12,000 records. The data has been pre-processed to ensure seamless rendering and interactivity in the browser.

2. Outline and Sketches

Our website is structured as a paginated data story. In the hero section, we introduce the topic with an animated sequence.

The Introduction

To make the introduction engaging, the cover page will feature a "Pulse of Life" SVG animation implemented with D3. A heartbeat line will sweep across the screen and smoothly transform into an upward trending line graph representing the global growth of life expectancy over time.

After a short introduction to the topic on the hero section, we move onto breaking down the specific factors that dictate human lifespan.

Question 1: Are there inherent biological or social differences in longevity?

To address this, our first visualization tackles the gender divide. We intend to use a Butterfly Chart where male life expectancy bars extend to the left and female life expectancy bars extend to the right. This allows for an immediate, intuitive comparison of the persistent survival gap on a global scale over the past two decades.



Figure 1: Butterfly Chart comparing male and female life expectancy

Question 2: How does geography dictate lifespan, and who is leading the race?

Moving beyond gender, we want to give users a macro level view of global health. We designed an interactive choropleth map. The user can drag a timeline slider to see how life expectancy has improved globally since the year 2000.

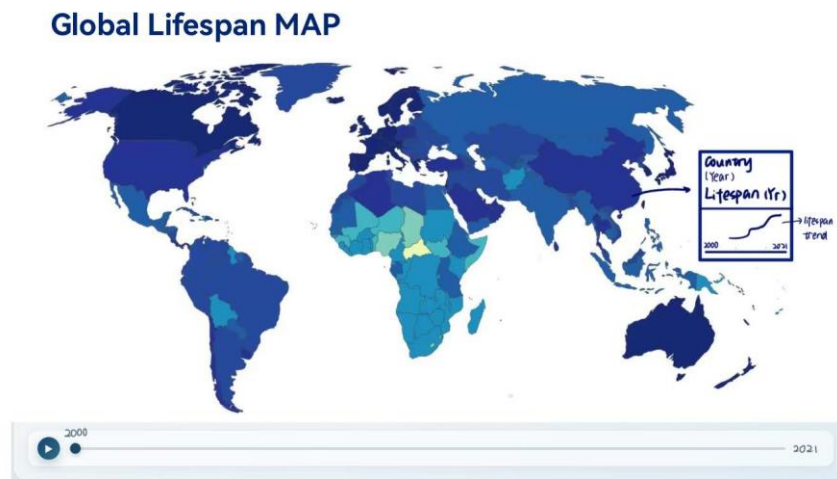


Figure 2: Interactive Choropleth Map with timeline slider

To complement the map, we want to highlight the nations pushing the boundaries of human survival. We will include a Racing Bar Chart that tracks the top 5 countries. When the user presses "Play", the bars will automatically reorder themselves year by year.

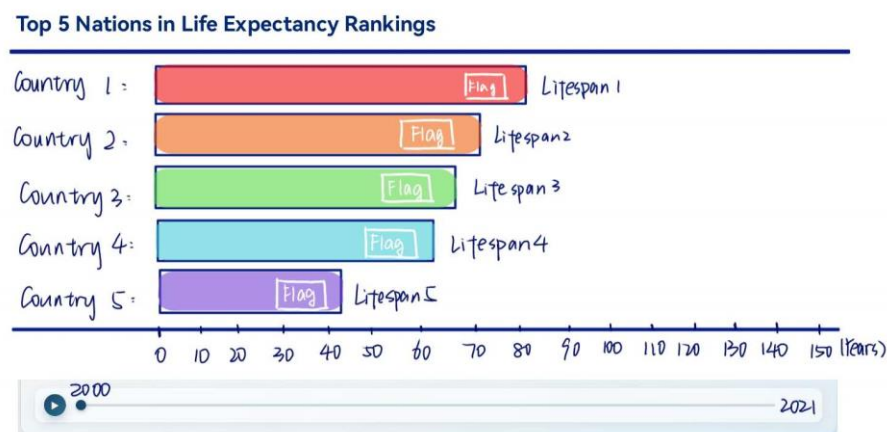


Figure 3: Animated Racing Bar Chart of Top 5 Nations

Question 3: Does more money always mean more time?

Here we investigate the "Wealth Paradox" through a combination of static comparisons and global context. We use The Bahamas as our primary anomaly to demonstrate that high national wealth does not automatically guarantee a long life.

- **Static Bar Comparison:** We first provide a direct bar chart comparison showing that despite a GDP per capita of over 30,000 dollars, Bahamians live nearly 10.4 years

shorter than their wealthy international peers.

- The Survival Filter (Scatter Plot):** To explain the reason behind this gap, we present a global scatter plot where the size of each bubble represents the Non-Communicable Disease (NCD) mortality rate. As seen in the figure 5, The Bahamas is a significant outlier. Its bubble is much larger than those of other nations in the same income bracket. This visualizes our core finding: the combination of extreme wealth inequality and a staggering 20.4 percent NCD mortality rate effectively neutralizes the benefits of a high GDP. This proves that longevity depends on how wealth is invested in health management and disease prevention.

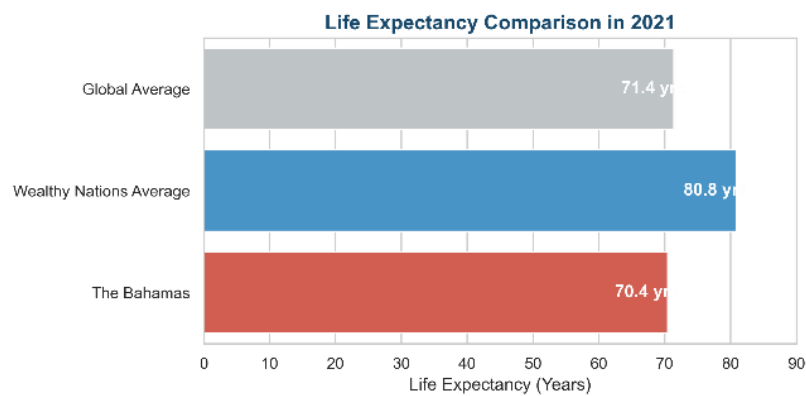


Figure 4: Static comparison chart highlighting The Bahamas anomaly

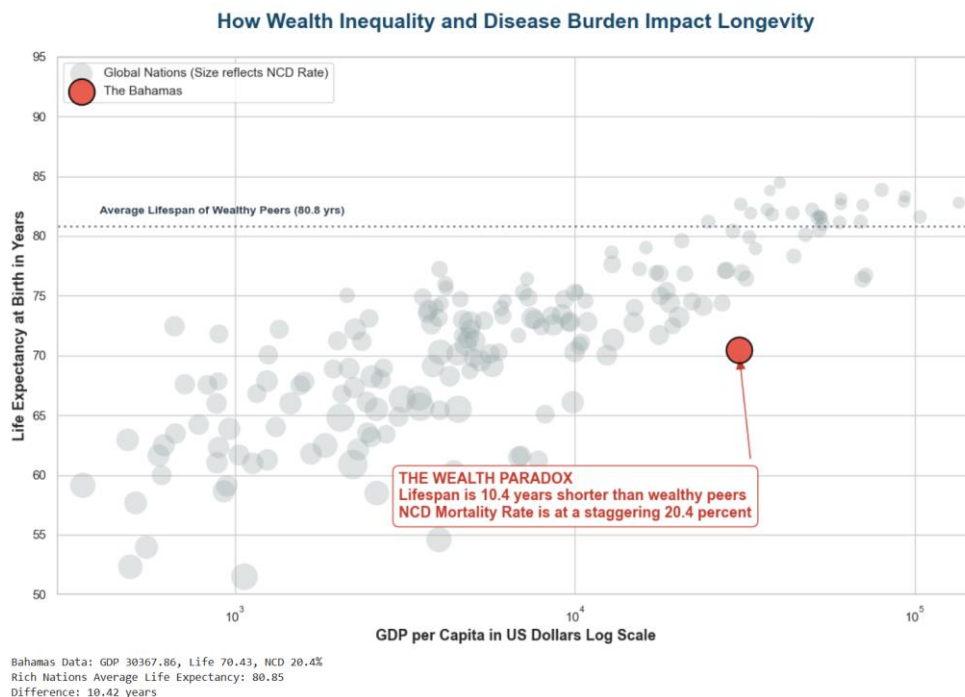


Figure 5. Analysis of the wealth paradox

Conclusion: How can we see the full picture for a single country?

Because measuring health is multidimensional, our final visualization is an interactive Radar Chart. Since the parameters (Wealth, Lifespan, Male Lifespan, Female Lifespan, NCD Rate)

have different units, the radar will display normalized relative changes. The user selects a specific country and uses a timeline slider to watch how that specific nation's "health footprint" has expanded or contracted from 2000 to 2021, comparing the country against its own historical baseline.

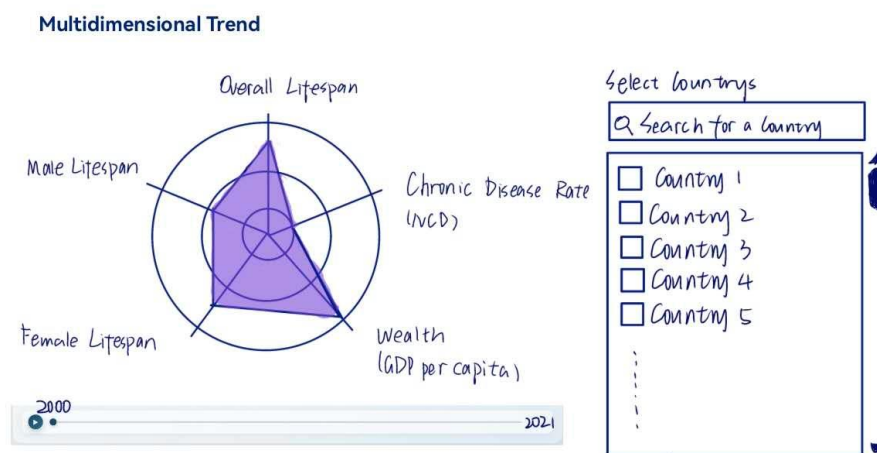


Figure 6: Interactive Radar Chart showing multidimensional changes over time

3. Tools and Lectures

We will build the website skeleton using HTML, CSS, and Vanilla JavaScript. All visualizations will be implemented using D3.js. The development is directly mapped to this course's syllabus:

- **The Web Layout:** Basic web development, Javascript Part 1 and 2.
- **Butterfly Chart:** D3.js plots, Perception colors.
- **Global Lifespan Map:** Practical maps, Maps, Interactions.
- **Racing Bar Chart:** Interactive D3, Storytelling.
- **Multidimensional Radar:** Tabular data, Mark and channels.

4. Extra Ideas

As long as it does not complicate the user experience, we would like to implement the following innovative ideas:

1. The first creative idea is adding a subtle heartbeat sound effect that plays during the cover page animation and speeds up or slows down based on the visual data.
2. Another challenging idea is adding a feature at the end of the story where the user inputs their own age, gender, and country. The visualization would then place a unique marker on the charts showing exactly where they stand compared to the global statistics.